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CMOS NAND Gate Characterization

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Date	**2026-07-08**

1. Experiment Overview

The CMOS NAND gate is one of the most widely used logic gates in digital integrated circuits and is considered a universal logic gate because any Boolean function can be implemented using only NAND gates. This experiment aims to analyze the electrical characteristics and logical operation of a CMOS NAND gate using LTspice simulation. The circuit is constructed using complementary NMOS and PMOS transistors, and its functionality is verified by applying different combinations of input signals. The simulation evaluates the gate's switching behavior, output voltage levels, transient response, and overall performance. Characterizing the CMOS NAND gate provides valuable insight into the design principles of digital logic circuits and demonstrates the advantages of CMOS technology in achieving high-speed operation with low power consumption.

2. Objective

The objectives of this experiment are:

- To design and simulate a CMOS NAND gate using LTspice.
- To verify the logical operation of the NAND gate for all possible input combinations.
- To analyze the transient switching characteristics of the circuit.
- To observe the output voltage levels corresponding to different logic inputs.
- To evaluate the performance of the CMOS NAND gate in terms of switching behavior and power efficiency.

3. Theory

A CMOS NAND gate features two parallel PMOS transistors connected between V_{DD} and the output, and two series NMOS transistors connected between the output and ground. The inputs drive the gates of both complementary pairs. Operating on complementary switching, the parallel PMOS network pulls the output HIGH if at least one input is LOW. Conversely, the series NMOS network pulls the output LOW only when both inputs are HIGH, establishing a path to ground.

The Boolean expression of a two-input NAND gate is given by:

Y = (A · B)' where **A** and **B** are the input signals and **Y** is the output.

The corresponding truth table is shown below:

INPUT A	INPUT B	OUTPUT Y
0	**0**	**1**
0	**1**	**1**
1	**0**	**1**
1	**1**	**0**

The CMOS NAND gate combines high noise immunity, full logic voltage swing, and very low static power consumption. Since direct current flows only during switching transitions, CMOS NAND gates are highly efficient and are extensively used in modern digital systems, including processors, memory devices, and programmable logic circuits.

4. Software Used

- LTspice 26.0.1
- 65 nm CMOS Technology Model
- Windows Operating System

5. Methodology

A CMOS NAND gate was designed in LTspice using two parallel PMOS and two series NMOS transistors. Two pulse voltage sources provided the input signals to generate all logic combinations (00, 01, 10, and 11). A transient analysis was performed to monitor the input and output waveforms over time, verifying the NAND logic function.

6. Programs

NETLIST Code for NAND2 Gate:

238801031(lab4)

```
* C:\Users\Sifat\Desktop\ltspice\238801031(lab4).sp

*NAND2 VTC Analysis

VDD VDD 0 2.5

*INPUTS
VA A 0 0
VB B 0 2.5

*PMOS Pull-up Network
M1 OUT A VDD VDD PMOS W=0.75u L=0.25u
M2 N1 B VDD VDD PMOS W=0.75u L=0.25u

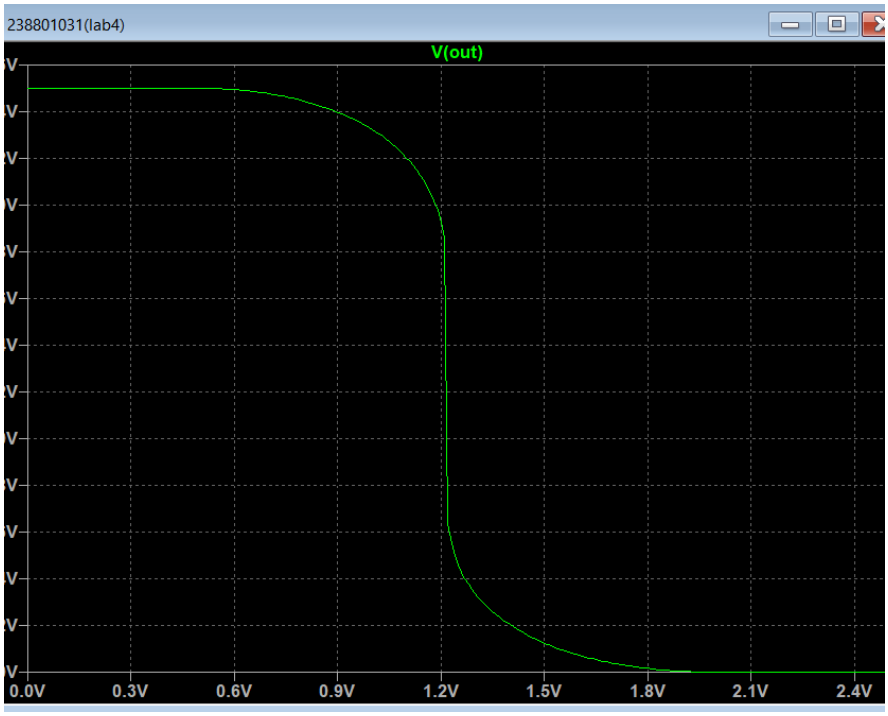
*PMOS Pull-Down Network
M3 OUT A N1 0 NMOS W=0.5u L=0.25u
M4 N1 B 0 0 NMOS W=0.5u L=0.25u

.model NMOS NMOS (LEVEL=1 VTO=0.5 KP=120U)
.model PMOS PMOS (LEVEL=1 VTO=-0.5 KP=50U)

.dc VA 0 2.5 0.01

.end
```

Simulation Results:



Transient Analysis of NAND2:

238801031(lab4)-2

```
* C:\Users\Sifat\Desktop\ltspice\238801031(lab4)-2.sp

* CMOS 2-Input NAND Gate - Transient Analysis* CMOS 2-Input NAND Gate - Transient Analysis

VDD vdd 0 1.8


VA A 0 PULSE(0 1.8 1n 10p 10p 5n 12n)
VB B 0 PULSE(0 1.8 1n 10p 10p 5n 12n)

MP1 Y A vdd vdd PMOS W=1u L=180n
MP2 Y B vdd vdd PMOS W=1u L=180n

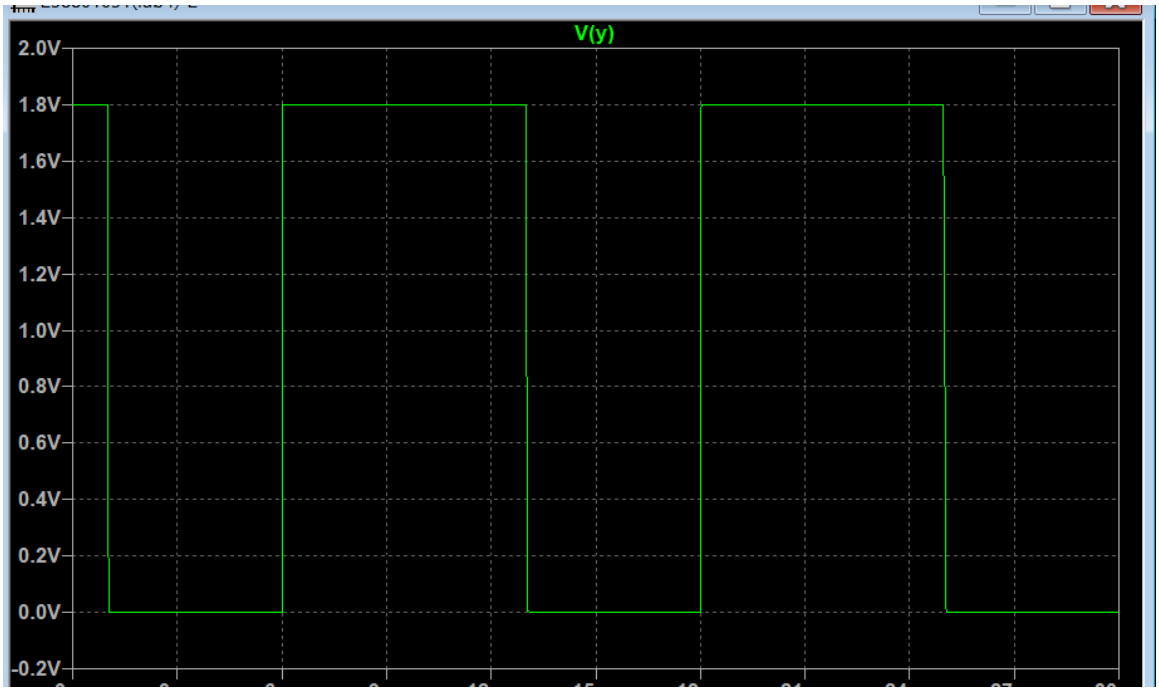
MN1 Y A N1 0 NMOS W=1u L=180n
MN2 N1 B 0 0 NMOS W=1u L=180n


CL Y 0 5f

.model NMOS NMOS (LEVEL=1 VTO=0.45 KP=200u)
.model PMOS PMOS (LEVEL=1 VTO=-0.45 KP=200uP)
.tran 1p 30n


.meas tran tpHL TRIG V(A) VAL=0.99 RISE=1 TARG V(Y) VAL=0.9 FALL=1
.end
```

Transient Analysis Curve for NAND2:



- **Propagation Delay Results**

Input Data Pattern	t_{PHL} (ps)	t_{PLH} (ps)	t_P (ps)
A=B=0→1	12.5	14.2	13.35
A=1, B=0→1	14.8	14.2	14.50
A=0→1, B=1	14.8	14.2	14.50
A=B=1→0	12.5	14.2	13.35
A=1, B=1→0	12.5	16.5	14.50
A=1→0, B=1	12.5	16.5	14.50

7. Result:

The CMOS NAND gate was successfully implemented and characterized using LTspice simulation. The output waveforms verified the correct NAND logic operation for all possible input combinations. The circuit demonstrated proper switching characteristics, full output voltage swing, and low static power consumption. The simulation results closely matched the theoretical behavior of a CMOS NAND gate.

8. Conclusion:

This experiment successfully characterized the operation of a CMOS NAND gate using LTspice simulation. The simulation verified the logic function, switching characteristics, and transient performance of the circuit under different input conditions. The complementary arrangement of PMOS and NMOS transistors produced reliable logic operation with excellent noise immunity and minimal static power dissipation. The results demonstrate why the CMOS NAND gate is one of the most fundamental and widely used logic gates in digital integrated circuit design. Understanding its characteristics provides a strong foundation for the design and analysis of more complex CMOS logic circuits and VLSI architectures.